Java-Based Test Automation Framework

Spring Boot · Selenium · REST-Assured · JDBC

Weeks 1–8 | Full Development Lifecycle

Objective

- Design a scalable Java-based test automation framework
- Integrate Selenium & REST-Assured for UI and API testing
- Enable parallel test execution with multithreading
- Provide real-time reporting and failure log collection
- Expose RESTful APIs via Spring Boot for test management
- Track and analyze test results with a built-in analytics module
- Deploy a fully functional, end-to-end testable application

Introduction

What It Is

- Enterprise-grade test automation platform
- Built with Java, Spring Boot & Maven/Gradle
- Centralized test execution and result storage
- JDBC-backed persistence for test entities
- RESTful API layer for external tool integration

Why It Matters

- Reduces manual testing overhead significantly
- Enables CI/CD pipeline integration
- Supports parallel execution for faster feedback
- Provides data-driven analytics on test trends
- Modular design ensures easy maintainability

Requirements & References

Language & Build

- Java (JDK 17+)
- Maven / Gradle
- Spring Boot 3.x

Testing Tools

- Selenium WebDriver
- REST-Assured
- JUnit / TestNG

Database & ORM

- JDBC (Spring Data)
- MySQL / PostgreSQL
- Spring Repositories

Execution & CI

- Multithreading (Java)
- Parallel Test Runs
- REST API Triggers

System Architecture

API Layer	*Spring Boot RESTful Controllers*
Service Layer	*Business Logic · Scheduling · Execution Engine*
Integration Layer	*Selenium · REST-Assured · Test Runners*
Data Layer	*JDBC Repositories · MySQL / PostgreSQL*
Analytics & Reporting	*Result Tracker · Report Generator · Log Collector*

▼ Data Flow ▼

Modules Overview

01

Test Integration Engine

- Selenium & REST-Assured setup
- JDBC-based data persistence
- Test entity model definitions

02

Scheduling & Execution

- Multithreaded parallel execution
- Test suite scheduling logic
- Spring Boot execution APIs

03

Reporting Hub

- Automated report generation
- Failure log aggregation
- Spring Boot reporting APIs

04

Result Analytics Tracker

- Trend analysis from stored data
- Analytics REST endpoints
- End-to-end validation support

Development Milestones

Week 1–2

Setup & Training

- JDK, DB, Spring Boot setup
- OOP, JDBC, Selenium training
- Test entity model definition

Week 3–4

Integration Engine

- Selenium/REST-Assured classes
- Spring Boot JDBC repos
- RESTful API development

Week 5–6

Scheduling & Reporting

- Multithreaded execution logic
- Report & log generation
- Scheduling & report APIs

Week 7–8

Analytics & Deploy

- Result analysis integration
- Analytics REST endpoints
- E2E testing & deployment

Screenshots / Outputs

SQL TABLE

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

id	test_case_id	status	error_message	execution_time	run_id
1	2	FAIL	Assertion failed	2026-04-12 19:55:47	7e60b8c9-0c7a-4048-a11e-572cf7b07c8f
2	4	PASS	NULL	2026-04-12 19:55:47	7e60b8c9-0c7a-4048-a11e-572cf7b07c8f
3	1	FAIL	Assertion failed	2026-04-12 19:55:47	7e60b8c9-0c7a-4048-a11e-572cf7b07c8f
4	5	FAIL	Assertion failed	2026-04-12 19:55:47	7e60b8c9-0c7a-4048-a11e-572cf7b07c8f
5	3	PASS	NULL	2026-04-12 19:55:47	7e60b8c9-0c7a-4048-a11e-572cf7b07c8f
6	4	FAIL	Assertion failed	2026-04-12 19:55:48	52e68676-66a7-4335-b162-a07da322a241
7	1	PASS	NULL	2026-04-12 19:55:48	52e68676-66a7-4335-b162-a07da322a241
8	2	FAIL	Assertion failed	2026-04-12 19:55:48	52e68676-66a7-4335-b162-a07da322a241
9	5	PASS	NULL	2026-04-12 19:55:48	52e68676-66a7-4335-b162-a07da322a241
10	3	PASS	NULL	2026-04-12 19:55:48	52e68676-66a7-4335-b162-a07da322a241
11	3	FAIL	Assertion failed	2026-04-12 19:56:47	44c73ac8-69c5-4b02-b54b-156f89473663
12	1	PASS	NULL	2026-04-12 19:56:47	44c73ac8-69c5-4b02-b54b-156f89473663
13	4	PASS	NULL	2026-04-12 19:56:47	44c73ac8-69c5-4b02-b54b-156f89473663
14	2	PASS	NULL	2026-04-12 19:56:47	44c73ac8-69c5-4b02-b54b-156f89473663
15	5	FAIL	Assertion failed	2026-04-12 19:56:47	44c73ac8-69c5-4b02-b54b-156f89473663

Run API

POST http://localhost:8080/api/execution/run Send

Docs | Params | Authorization | Headers 8 | Body | Scripts | Tests | Settings | Cookies

Query Params

Key	Value	Description	Bulk Edit
Key	Value	Description	

Body 2:37 AM 200 OK 263 ms 196 B

Raw | Preview | Visualize

1 Execution Completed Successfully

Results

localhost:8080/api/execution/results

retty-print


```
{
  "id": 1,
  "testCaseId": 2,
  "status": "FAIL",
  "errorMessage": "Assertion failed",
  "executionTime": "2026-04-13T01:25:47",
  "runId": "7e60b8c9-0c7a-4048-a11e-572cf7b07c8f"
},
{
  "id": 2,
  "testCaseId": 4,
  "status": "PASS",
  "errorMessage": null,
  "executionTime": "2026-04-13T01:25:47",
  "runId": "7e60b8c9-0c7a-4048-a11e-572cf7b07c8f"
},
{
  "id": 3,
  "testCaseId": 1,
  "status": "FAIL",
  "errorMessage": "Assertion failed",
  "executionTime": "2026-04-13T01:25:47",
  "runId": "7e60b8c9-0c7a-4048-a11e-572cf7b07c8f"
},
{
  "id": 4,
  "testCaseId": 5,
  "status": "FAIL",
  "errorMessage": "Assertion failed",
  "executionTime": "2026-04-13T01:25:47",
  "runId": "7e60b8c9-0c7a-4048-a11e-572cf7b07c8f"
},
{
  "id": 5,
  "testCaseId": 3,
```


Summary

localhost:8080/api/execution/summary

TOTAL: 740 | PASS: 507 | FAIL: 233 | PASS%: 68.51

Key Technical Outcomes

Achievements

- Fully modular 4-component architecture
- RESTful APIs for all major operations
- Real-time parallel test execution
- Automated failure detection & logging
- End-to-end test coverage validated

Skills Demonstrated

- Java OOP & design patterns
- Spring Boot & REST API development
- Selenium & REST-Assured integration
- JDBC data modeling & persistence
- Multithreading & scheduling concepts

Conclusion

- A robust, production-ready test automation framework built in Java
- Spring Boot APIs enable seamless CI/CD pipeline integration
- Parallel execution and scheduling reduce testing cycle times
- Analytics module provides actionable insights on test quality
- All 4 modules delivered across 8 weeks with full deployment
- Demonstrates real-world enterprise Java development practices

Java · Spring Boot · Selenium · REST-Assured · JDBC · Analytics

THANK YOU

Any queries?

Java · Spring Boot · Selenium · REST-Assured · JDBC · Analytics